WEEK 1 · 9–12 NOVEMBER 2026

Data access, visualisation and exploration

From a public catalogue to a spectrum on your screen

Week 1 · Session 4 · practical: 00_EO_data_quickstart_notebook.ipynb

LEARNING OBJECTIVES

What you will be able to do

- Navigate a STAC catalogue: collection, item, asset
- Say what is inside a Tanager HDF5 file and find the wavelengths
- Preview a scene cheaply before downloading a gigabyte
- Read a window or a subset of bands instead of the whole cube
- Choose a scene for your own project using real availability

1. Finding data

STAC, the industry standard catalogue

STAC in one slide

- **SpatioTemporal Asset Catalog** — a JSON convention for describing Earth observation data
- **Catalog** → the top level, links to collections
- **Collection** → a themed group of scenes, e.g. `fire`, `urban`, `GHG-plumes`
- **Item** → one acquisition: geometry, datetime, cloud cover, plus metadata
- **Asset** → one downloadable file belonging to that item
  - Planet's Tanager catalogue is *static*: plain JSON files, no server, no account, no API key

Fetching one scene's metadata

```
import requests

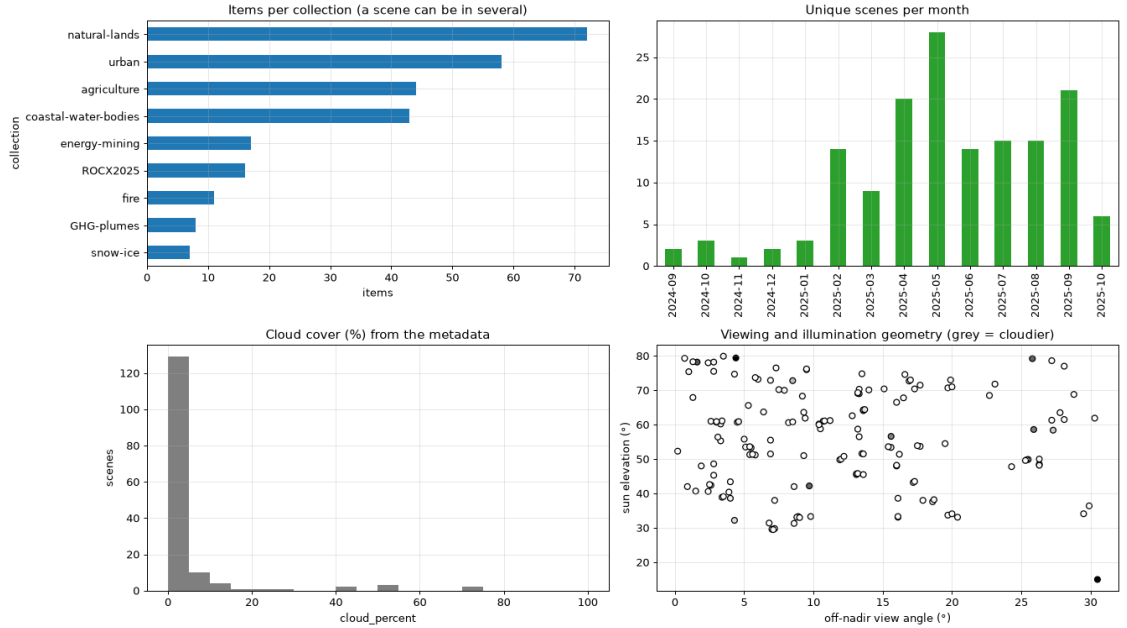
base = "https://www.planet.com/data/stac/tanager-core-imagery"
item = requests.get(f"{base}/agriculture/20250608_091605_90_4001/"
                    "20250608_091605_90_4001.json").json()

print(item["properties"]["location_description"])   # Çumra, Konya, Türkiye
print(item["properties"]["cloud_percent"])          # 2
print(sorted(item["assets"]))                       # what you can download
```

No authentication, no SDK — the catalogue is just JSON over HTTPS.

What is actually in the archive

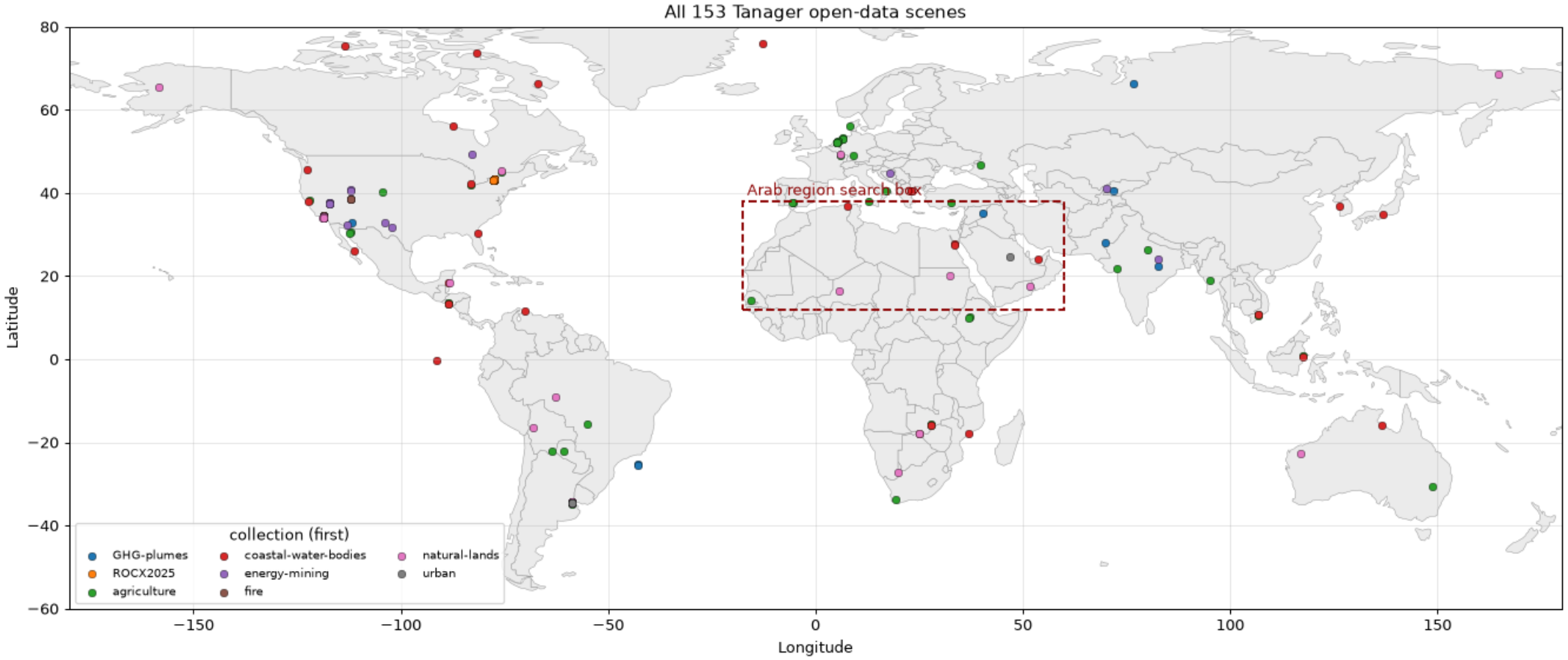
- **153 unique scenes**, from 25 Sep 2024 to 29 Oct 2025
- 276 catalogue entries — 123 scenes appear in more than one collection
- Nine themed collections, from `agriculture` to `snow-ice`
- Scene footprints span 322–879 km², median **394 km²**



Tanager open data by collection and over time

GLOBAL COVERAGE IS SPARSE

Where the scenes are



Footprint centroids, coloured by collection

2. Inside a Tanager scene

One HDF5 file, many layers

What the surface reflectance file contains

Layer	Shape	What it is for
surface_reflectance	426 × rows × cols	The data cube itself
surface_reflectance_uncertainty	426 × rows × cols	Per-pixel, per-band uncertainty → SNR
nodata_pixels	rows × cols	Outside the swath after orthorectification
beta_cloud_mask / beta_cirrus_mask	rows × cols	Cloud screening — always inspect it
column_water_vapour, aerosol_optical_depth	rows × cols	Atmospheric correction inputs
sun / sensor zenith and azimuth	rows × cols	Illumination and viewing geometry

Where the wavelengths live

- Attached to the `surface_reflectance` dataset as attributes, not as a separate file
- **wavelengths** — 426 band centres in nanometres (376–2499 nm)
- **fwhm** — the width of each band (5.2–6.8 nm)
- **good_wavelengths** — a flag marking bands the provider considers unusable
- The grid (CRS, corner coordinates, pixel size) is in `HDFEOS INFORMATION/StructMetadata.0`
  - The STAC item also carries wavelengths, as `eo:center_wavelength` in micrometres

The assets, and when to use each

Asset	Size	Use it for
thumbnail	kilobytes	Deciding whether a scene is worth a closer look
ortho_visual	megabytes	A true-colour COG you can stream into a map
ortho_sr_hdf5	~1 GB	The surface reflectance cube — the real analysis product
ortho_radiance_hdf5	~1 GB	At-sensor radiance, needed for gas retrievals
ortho_beta_udm	megabytes	Usable-data mask as a GeoTIFF

THE WORKFLOW RULE OF THE DAY

Never download a gigabyte to find out the scene is cloudy.

The thumbnail is a few kilobytes and the visual COG streams. Look first, download second — and when you do download, read only the window and the bands you need.

3. Reading efficiently

Windows, strides and bands

Three ways to read less

```
import hsi_utils as hu
scene = hu.get_scene("20250608_091605_90_4001")   # local file, or downloads it

# 1. a spatial window: (row_start, row_stop, col_start, col_stop)
cube, wl = scene.read_cube(window=(200, 400, 300, 500))

# 2. a strided preview of the whole scene — every 4th row and column
preview, wl = scene.read_cube(stride=4)

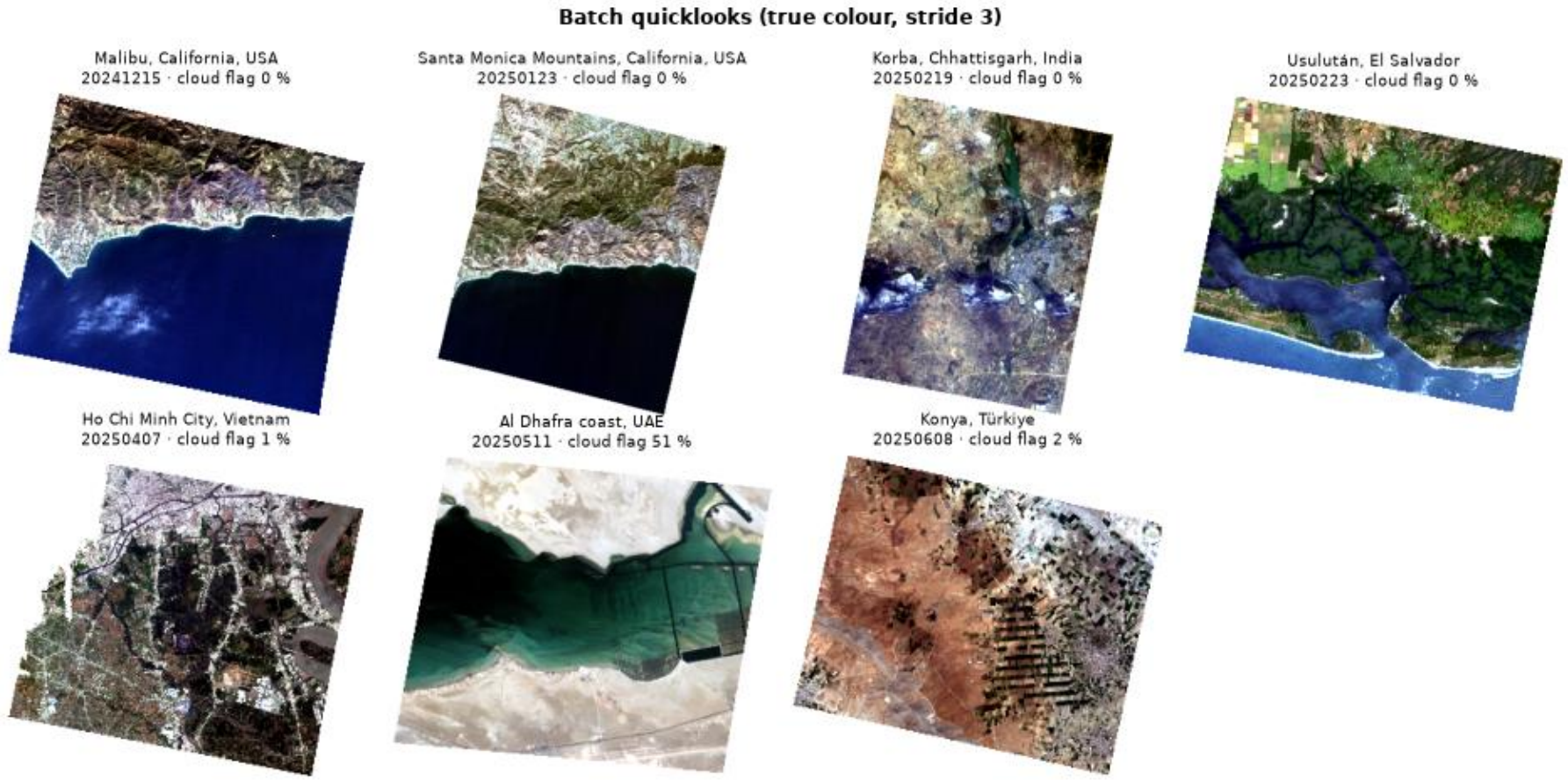
# 3. one band, by wavelength in nm
nir = scene.read_band(860)
```

A full cube is ~950 MB in memory; a 200 × 200 window is about 55 MB.

Two habits worth forming now

- **Preview with a stride, analyse in a window.** Full-resolution full-scene reads are rarely needed
- **Check the masks before trusting any number** — no-data, cloud and cirrus
  - Around 20–30 % of every Tanager grid is no-data: the swath is rotated inside a north-up grid
- Invalid pixels come back as NaN, so use `np.nanmean` and friends throughout
- If your statistic looks strange, map the mask first — it is usually the explanation

Six scenes, six very different problems



True-colour quicklooks of the course scenes (stride 3)

4. Choosing a scene for your project

Availability decides feasibility

What to filter on

- **Cloud cover** — `cloud_percent` in the item properties, but verify with the thumbnail
- **Off-nadir angle** — large angles distort geometry and change illumination
- **Sun elevation** — low sun means long shadows, a problem in terrain
- **Date** — does it match the phenology or the event you care about?
- **Repeat coverage** — change detection needs two scenes over the same place
  - Six Tanager scenes cover Malibu, which is why the fire change-detection example exists

Availability over the Arab region

Query	Result
Scenes intersecting the region box	14
Of those, in Arab League countries	8
Countries represented	Syria, Sudan, UAE, Saudi Arabia, Algeria, Egypt
Remaining after a ≤ 20 % cloud filter	5

PRACTICAL

Over to you

▶ `00_EO_data_quickstart_notebook.ipynb` · `120 min`

- Search a STAC catalogue and inspect an item's properties and assets
- Display a thumbnail and stream a visual COG onto an interactive map
- Open the HDF5, read the wavelengths, and plot a full 426-band spectrum
- Compute your first indices and check them against the quality masks
- Then pick one theme notebook (01–06) that matches your interest and run it

KEY TAKEAWAYS

Recap

- STAC: catalog → collection → item → asset; Tanager's is static, open and needs no account
- The HDF5 holds the cube, the masks, the uncertainty and the **wavelengths as attributes**
- Preview with thumbnails and strides; download and read full resolution only when you must
- Always inspect the masks before trusting a statistic
- **Check availability before committing to an area** — notebook 16 does this in one cell

Next: Session 5 — Preprocessing concepts for hyperspectral data

References and further reading

STAC specification — stacspec.org

Planet Tanager open data catalogue (CC-BY-4.0) — www.planet.com/data/stac

Planet Tanager product documentation — docs.planet.com/data/imagery/tanager

Course notebooks: 00_EO_data_quickstart_notebook.ipynb, 16_scene_finder.ipynb

Scoping canvas — templates/project_scoping_template.md